

➤ Introduction and Installation of Operating System:

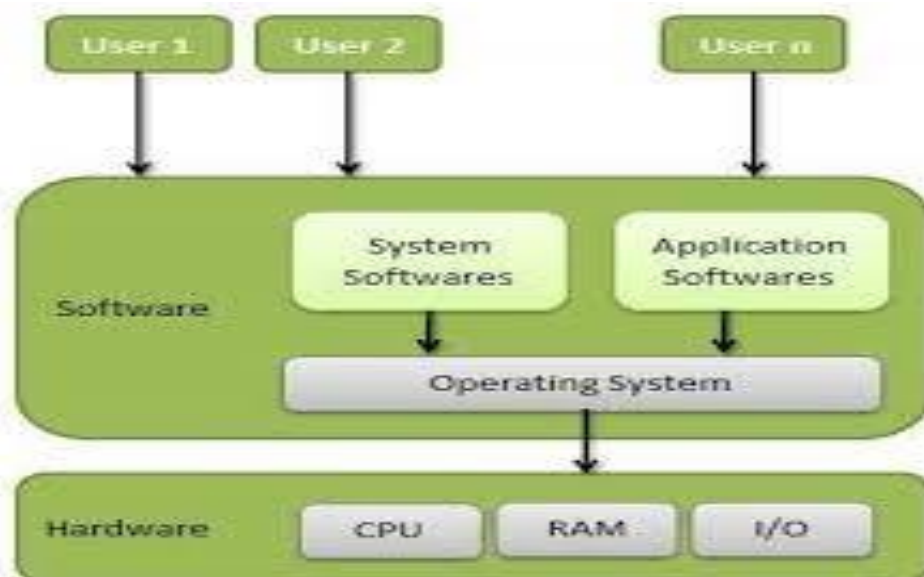
❖ Introduction of Operating System

An Operating System (OS) is an interface between computer user and computer hardware. An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers.

Some popular Operating Systems include Linux Operating System, Windows Operating System, VMS, OS/400, AIX, z/OS, etc.

Definition

An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.



❖ Functionality of Operating System

Following are some of important functions of an operating System.

- Memory Management
- Processor Management
- Device Management
- File Management
- Security
- Control over system performance
- Job accounting
- Error detecting aids
- Coordination between other software and users

1. Security–

The operating system uses password protection to protect user data and similar other techniques. it also prevents unauthorized access to programs and user data.

2. Control over system performance –

Monitors overall system health to help improve performance. Records the response time between service requests and system response to have a complete view of the system health. This can help improve performance by providing important information needed to troubleshoot problems.

3. Job accounting –

Operating system Keeps track of time and resources used by various tasks and users, this information can be used to track resource usage for a particular user or group of user.

4. Error detecting aids –

Operating system constantly monitors the system to detect errors and avoid the Malfunctioning of computer system.

5. Coordination between other software and users –

Operating systems also coordinate and assign interpreters, compilers, assemblers and other software to the various users of the computer systems.

6. Memory Management –

Memory management refers to management of Primary Memory or Main Memory. Main memory is a large array of words or bytes where each word or byte has its own address.

Main memory provides a fast storage that can be accessed Directly by the CPU. For a program to be executed, it must in the main memory. An Operating System does the following activities for memory management –

- Keeps tracks of primary memory, i.e., what part of it are in use by whom, what part are not in use.
- In multiprogramming, the OS decides which process will get memory when and how much.
- Allocates the memory when a process requests it to do so.
- De-allocates the memory when a process no longer needs it or has been terminated.

7. Processor Management –

In multiprogramming environment, the OS decides which process gets the processor when and for how much time. This function is called process scheduling. An Operating System does the following activities for processor management –

- Keeps tracks of processor and status of process. The program responsible for this task is known as traffic controller.
- Allocates the processor (CPU) to a process.
- De-allocates processor when a process is no longer required.

8. Device Management –

An Operating System manages device communication via their respective drivers.

It does the following activities for device management –

- Keeps tracks of all devices. Program responsible for this task is known as the I/O controller.
- Decides which process gets the device when and for how much time.
- Allocates the device in the efficient way.
- De-allocates devices.

9. File Management –

A file system is organized into directories for efficient or easy navigation and usage. These directories may contain other directories and other files. An Operating System carries out the following file management activities. It keeps track of where information is stored, user access settings and status of every file and more... These facilities are collectively known as the file system.

❖ Types of Operating Systems:

Some of the widely used operating systems are as follows-

1. Batch Operating System –

This type of operating system does not interact with the computer directly. There is an operator which takes similar jobs having same requirement and group them into batches. It is the responsibility of operator to sort the jobs with similar needs.

Advantages of Batch Operating System:

- It is very difficult to guess or know the time required by any job to complete. Processors of the batch systems know how long the job would be when it is in queue
- Multiple users can share the batch systems
- The idle time for batch system is very less
- It is easy to manage large work repeatedly in batch systems

Disadvantages of Batch Operating System:

- The computer operators should be well known with batch systems
- Batch systems are hard to debug
- It is sometime costly
- The other jobs will have to wait for an unknown time if any job fails

2. Time-Sharing Operating Systems –

Each task is given some time to execute, so that all the tasks work smoothly. Each user gets time of CPU as they use single system. These systems are also known as Multitasking Systems. The task can be from single user or from different users also. The time that each task gets to execute is called quantum. After this time interval is over OS switches over to next task.

Advantages of Time-Sharing OS:

- Each task gets an equal opportunity
- Less chances of duplication of software
- CPU idle time can be reduced

Disadvantages of Time-Sharing OS:

- Reliability problem □ One must have to take care of security and integrity of user programs and data
- Data communication problem

3. Distributed Operating System –

These types of operating system is a recent advancement in the world of computer technology and are being widely accepted all-over the world and, that too, with a great pace. Various autonomous interconnected computers communicate each other using a shared communication network. Independent systems possess their own memory unit and CPU.

Advantages of Distributed Operating System:

- Failure of one will not affect the other network communication, as all systems are independent from each other
- Electronic mail increases the data exchange speed
- Since resources are being shared, computation is highly fast and durable
- Load on host computer reduces
- These systems are easily scalable as many systems can be easily added to the network
- Delay in data processing reduces

Disadvantages of Distributed Operating System:

- Failure of the main network will stop the entire communication
- To establish distributed systems the language which are used are not well defined yet

- These types of systems are not readily available as they are very expensive. Not only that the underlying software is highly complex and not understood well yet.

4. Network Operating System –

These systems run on a server and provide the capability to manage data, users, groups, security, applications, and other networking functions. These type of operating systems allow shared access of files, printers, security, applications, and other networking functions over a small private network. One more important aspect of Network Operating Systems is that all the users are well aware of the underlying configuration, of all other users within the network, their individual connections etc.

Advantages of Network Operating System:

- Highly stable centralized servers
- Security concerns are handled through servers
- New technologies and hardware up-gradation are easily integrated to the system
- Server access are possible remotely from different locations and types of systems

Disadvantages of Network Operating System:

- Servers are costly
- User has to depend on central location for most operations
- Maintenance and updates are required regularly

5. Real-Time Operating System –

These types of OSs serves the real-time systems. The time interval required to process and respond to inputs is very small. This time interval is called **response time**.

Two types of Real-Time Operating System which are as follows:

- **Hard Real-Time Systems:**

These OSs are meant for the applications where time constraints are very strict and even the shortest possible delay is not acceptable. These systems are built for saving life like automatic parachutes or air bags which are required to be readily available in case of any accident. Virtual memory is almost never found in these systems.

- **Soft Real-Time Systems:**

These OSs are for applications where for time-constraint is less strict.

❖ **Key Features of Various Operating System (OS)**

An Operating System provides essential features that enable efficient, secure, and reliable functioning of a computer system.

1. Protected Mode and Supervisor Mode

- The OS operates in **protected/supervisor mode** to control access to hardware and critical system resources.
- Prevents user programs from directly accessing hardware.
- Ensures system stability and security.

Example: User applications cannot directly access CPU registers or memory areas reserved for the OS.

2. Disk Access and File System Support

- The OS provides access to storage devices like hard disks and SSDs.
- Manages **file systems** to organize data into files and directories.
- Supports file operations such as create, read, write, delete, and rename.

3. Device Drivers

- Device drivers act as an interface between hardware devices and the OS.
- Enable communication with peripherals like printers, scanners, keyboards, and displays.
- Allow hardware independence.

4. Networking

- Provides networking capabilities for data communication.
- Supports protocols like TCP/IP.
- Enables resource sharing, internet access, and remote communication.

Example: File sharing over LAN or accessing websites.

5. Security

- Protects system and user data from unauthorized access.
- Uses authentication (username, password), authorization, and encryption.
- Prevents malware and unauthorized modifications.

6. Program Execution

- Loads programs into memory and executes them.
- Handles process creation, scheduling, and termination.
- Ensures smooth execution of multiple programs.

7. Memory Management

- Allocates and deallocates memory to processes.
- Tracks memory usage and prevents conflicts.
- Improves system performance and efficiency.

8. Virtual Memory

- Uses disk space as an extension of RAM.
- Allows execution of large programs even with limited physical memory.
- Improves multitasking capability.

❖ Hardware & Software Requirements for Operating System installation

- The hardware and software requirements for operating system installation can vary depending on the specific operating system you want to install. Below, I'll provide a general overview of the typical requirements you might encounter when installing a modern desktop operating system, such as Windows, macOS, or a Linux distribution:

Hardware Requirements

Processor (CPU): The CPU requirements can vary significantly based on the operating system. For most modern operating systems, a 64-bit multi-core processor is recommended. Specific CPU requirements may be listed by the OS vendor.

Memory (RAM): Typically, a minimum of 2 GB of RAM is required for basic functionality, but 4 GB or more is recommended for a smoother experience, especially for multitasking or resource-intensive applications.

Storage: You'll need storage space for the operating system, applications, and your files. The minimum storage space required can range from 16 GB to 32 GB, but it's recommended to have more, especially if you plan to install a lot of software or store a lot of data. Solid-state drives (SSDs) are preferable for better performance.

Graphics Card (GPU): Many modern operating systems can run with integrated graphics, but if you plan to use graphics-intensive applications or gaming, you may need a dedicated GPU. Check the specific requirements for your use case.

Monitor: A monitor or display device compatible with your chosen operating system. Check the recommended resolution and refresh rate settings.

Input Devices: A keyboard and a pointing device, such as a mouse or touchpad.

Network Connectivity: An Ethernet or Wi-Fi adapter to connect to the internet or local network. Drivers for your network hardware may be required.

Optical Drive (Optional): Some systems may still require a CD/DVD drive for installation, although this is becoming less common as operating systems are often installed from USB drives or over the network.

Software Requirements

Installation Media: You'll need a bootable installation medium, such as a USB flash drive or a DVD with the operating system. You can create installation media using tools provided by the OS vendor.

Product Key or License: For commercial operating systems like Windows, you'll need a valid product key or license to activate and use the OS. Linux distributions are typically open-source and do not require a product key.

Drivers: Some hardware components may require specific drivers to work correctly with the operating system. It's a good idea to check for driver compatibility with your hardware before installation.

Internet Connection (Optional): An internet connection can be useful during the installation process for downloading updates, drivers, or additional software. It's not always required during the installation but is recommended.

Always consult the official documentation provided by the operating system vendor for the most accurate and up-to-date hardware and software requirements for the specific OS version you plan to install. Additionally, consider your intended use case when determining

whether your hardware meets the minimum or recommended requirements.

❖ The Steps To Install Various Operating Systems

- The steps to install various operating systems can vary depending on the specific OS and the hardware you're using. I'll provide a general overview of the installation process for some common operating systems: Windows, macOS, and Linux.

1. Windows:

Obtain a Windows installation media, usually in the form of a bootable USB drive or a DVD.

Insert the installation media into your computer and restart it.

Boot from the installation media by accessing the boot order to prioritize the installation media.

Follow the on-screen instructions to select the language, region, and keyboard layout.

Enter the product key if required.

Choose the installation type (typically "Custom" for a clean install or "Upgrade" to keep your files and settings).

Select the partition or drive where you want to install Windows and format it if necessary.

Follow the prompts to complete the installation, which includes setting up your user account and network settings.

2. MacOS:

Download macOS from the Mac App Store or obtain a macOS installation disk or USB drive.

Insert the installation media and restart your Mac.

Hold down the Option (Alt) key during startup to access the boot menu and select the installation media.

Choose the "Install macOS" option from the menu.

Follow the on-screen instructions to select the destination drive for the installation.

Sign in with your Apple ID if required.

Customize your installation preferences (e.g., language, keyboard layout, network settings).

Wait for the installation to complete, and follow the setup assistant to configure your Mac.

3. **Linux** (using Ubuntu as an example):

Download the desired Linux distribution (e.g., Ubuntu) and create a bootable USB drive using tools like Rufus (Windows) or Etcher (Windows, macOS, Linux).

Insert the bootable USB drive into your computer and restart it.

Access the boot menu or BIOS/UEFI Basic Input/output System (BIOS) or the Unified Extensible Firmware Interface (UEFI) settings to set the USB drive as the primary boot device.

Select "Install Ubuntu" from the boot menu.

Choose your language and keyboard layout.

Select the installation type, such as "Erase disk and install Ubuntu" or "Something else" for manual partitioning.

Create or select the partition where you want to install Linux.

Set your location and create a user account. Follow the installation wizard to configure additional settings, such as time zone and software selection.

Wait for the installation to complete, and then remove the USB drive when prompted.

Keep in mind that these are general guidelines, and the exact steps may vary based on the specific version of the operating system and your hardware configuration. Always refer to the official

documentation provided by the OS developer for the most accurate and up-to-date instructions.

❖ **Installation of operating system through ghost image introduction of Linux**

Installing an operating system, such as Linux, using a ghost image (also known as a disk image or clone image) involves creating a copy of an existing system's disk or partition and then deploying that image to one or more target systems. Ghost imaging is a common method for efficiently deploying a consistent OS setup to multiple computers. Here's a general overview of the process:
Here are the general steps:

- **Prepare the Source System:**

Set up a reference system with the Linux distribution you want to clone.

Install and configure all the software and settings you want to include in the image.

Ensure the source system is in a clean and stable state.

- **Create a Disk Image:**

Use the ghost imaging software to create a disk image of the source system. This involves taking a snapshot of the entire disk or specific partitions.

Save the disk image to an external storage device or network location.

- **Prepare the Target System:**

Ensure that the target system is prepared for imaging. It should have the same or similar hardware as the source system.

Boot the target system using a bootable media or network boot method, depending on your imaging software.

- **Deploy the Ghost Image:**

On the target system, use the ghost imaging software to restore the disk image to the target system's disk or partition. This process replaces the target system's data with the contents of the image.

Make sure the imaging tool provides options for configuring partition sizes and disk layout, as these may differ between source and target systems.

- **Customize Configuration (Optional):**

After deploying the image, you might need to customize system-specific settings, such as hostname, IP address, or other unique configurations.

- **Test and Validate:**

Boot the target system to ensure the Linux installation is functional and all configurations are in place.

- **Repeat for Multiple Systems:**

If you need to deploy the same image to multiple systems, repeat the process for each target system.

Ghost imaging is particularly useful for system administrators or IT professionals who need to set up multiple systems with the same software configuration quickly. However, it's important to be cautious when using this approach, as it may not account for hardware differences between systems. Ensure that the hardware of the target systems is sufficiently similar to the source system to avoid compatibility issues. Also, be aware of licensing and activation issues with proprietary software when using ghost imaging.

❖ Introduction of Linux

Linux is a free, open-source operating system known for its flexibility, stability, and strong security. It is widely used in personal computing, server environments, and enterprise systems because of its performance and customization capabilities.

- Offers high security and stability, making it ideal for servers and development work.
- Fully open-source and free to use, modify, and distribute.
- Highly flexible and customizable to suit different user and industry needs.

- Supported by a large global community and a vast software ecosystem.

Linux consists of:

- Kernel (**core part**)
- Shell
- System Libraries
- Utilities & Applications

❖ Computer Booting Process, and installation of drivers.

- The computer booting process is a series of steps that occur when you turn on your computer. It involves loading the operating system, initializing hardware components, and preparing the computer for use. Additionally, the installation of drivers is a crucial step in ensuring that the computer's hardware devices work properly. Here's an overview of both processes:

- **Booting Process:**

Power On: When you press the power button, the computer's power supply provides electricity to the components, and the CPU (Central Processing Unit) begins execution.

BIOS/UEFI: The Basic Input/output System (BIOS) or the Unified Extensible Firmware Interface (UEFI) initializes the hardware components. This includes checking for connected devices, such as hard drives and RAM, and ensuring they are functional.

Boot Loader: The boot loader (e.g., GRUB for Linux or Windows Boot Manager for Windows) takes control. It displays a menu if you have multiple operating systems installed and allows you to select one.

Operating System Kernel: The selected operating system's kernel is loaded into memory. This is the core of the operating system and manages hardware and resources.

Initialization and User Login: The operating system initializes hardware drivers and services, configures network settings, and

prepares for user login. You'll see the login screen or desktop environment depending on your system.

Installation of Drivers:

Hardware Detection: When the operating system loads, it attempts to identify and detect the hardware components in your computer. It may recognize some devices without the need for additional drivers.

Driver Search: If the operating system doesn't have built-in drivers for certain hardware, it will search for suitable drivers. These drivers are typically provided by the hardware manufacturer and are included with the operating system or available for download.

Driver Installation:

Automatic Installation: In many cases, the operating system will automatically install the necessary drivers from its driver repository.

Manual Installation: If the automatic installation fails or if you want to update a driver, you can manually install drivers by downloading them from the manufacturer's website and running the installation program.

Driver Updates: It's keep your drivers up to date. You can periodically check for driver updates on the manufacturer's website or use driver update software to ensure your hardware is running with the latest enhancements and fixes.

Driver Verification: After installation, the operating system verifies that the drivers are compatible and functional. If there are issues, it may prompt you to resolve them.

Restart: In some cases, a system restart is required to complete the driver installation process. Properly installing and updating drivers is crucial for ensuring your computer's hardware components work correctly and efficiently. Outdated or incompatible drivers can lead to hardware issues and system instability.